

ONLINE MASTERS IN **DATA SCIENCE**

DSC 257R - UNSUPERVISED LEARNING

# THE BAYES NET FORMALISM

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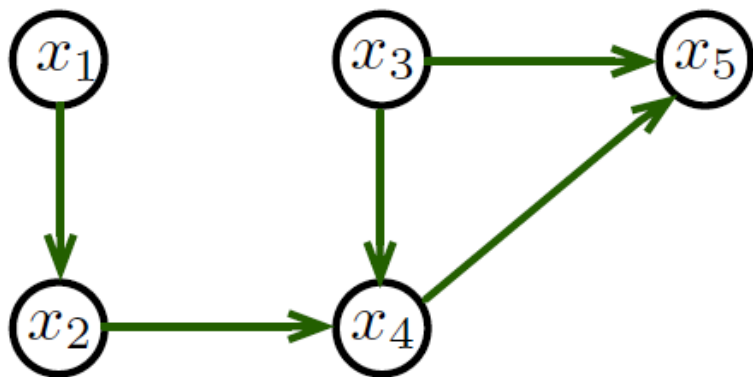
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## Bayes Net Formalism

Let  $X_1, \dots, X_n$  be a collection of random variables.

- Let  $G$  be a **directed acyclic graph** (DAG) with nodes named  $X_1, \dots, X_n$ .
- Let  $\Pi_i$  denote the parents of  $X_i$  in  $G$ .



If  $P$  is a distribution over  $(X_1, \dots, X_n)$ , we say  $P$  **factors over**  $G$  if

$$P(X_1, \dots, X_n) = P(X_1|\Pi_1)P(X_2|\Pi_2) \cdots P(X_n|\Pi_n)$$

## Bayes Net Formalism (Cont'd)

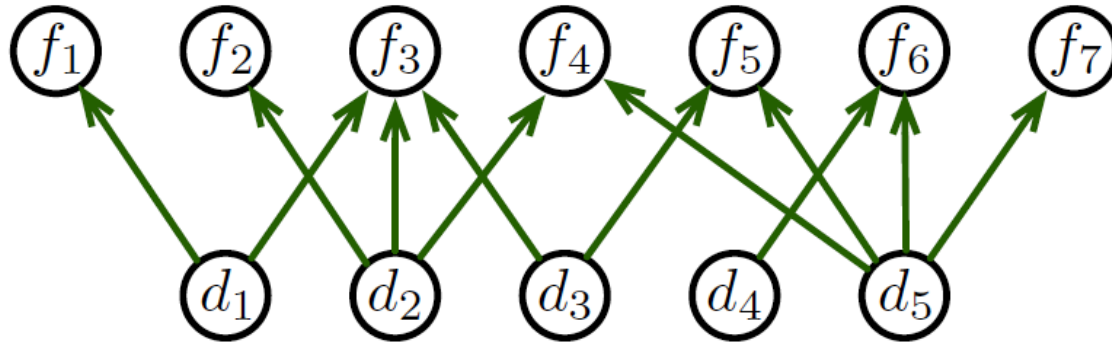
A DAG  $G$  represents the family of probability distributions that factor over  $G$ .

- 1 Is there a DAG that captures *all* probability distributions over the  $n$  variables?
- 2 If the  $X_i$  take on finitely many values, what additional information is needed to specify a probability distribution over  $G$ ?

## Example: QMT-DT Network

Shwe et al (1991). *Probabilistic Diagnosis Using a Reformulation of the INTERNIST-1/QMR Knowledge Base.*

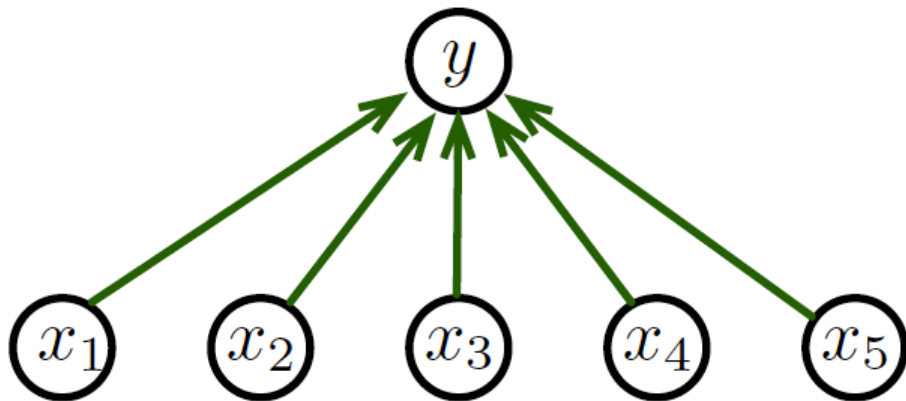
A Bayes net over more than 600 diseases and 4000 symptoms (findings).



$$\Pr(d, f) = \prod_i \Pr(d_i) \prod_j \Pr(f_j | \text{parents in } d)$$

## Example (Cont'd): Noisy-OR Conditional Probabilities

Generalization of  $Y = X_1 \vee X_2 \vee X_3 \vee \dots$



$$\Pr(Y = 0 | X_1, X_2, \dots) = (1 - \lambda) \prod_i (1 - p_i)^{X_i}.$$